

QUADRATIC EQUATIONS

Chapter

4



Exercise-1

Part - 1 (Basic Multiple Choice Questions)

- Which of the following satisfy the equation $a^2b^2x^2 + b^2x - a^2x - 1 = 0$
(a) $\frac{1}{a^2}$ (b) $\frac{1}{b^2}$ (c) $\frac{-1}{b^2}$ (d) None of these
- The roots of the quadratic equation $x^2 - 0.04 = 0$ are
(a) ± 0.2 (b) ± 0.02 (c) 0.4 (d) 2
- The root(s) of the equation $\sqrt{2x^2 + 9} = 9$ is / are :
(a) $x = 6$ (b) $x = \pm 6$ (c) $x = -6$ (d) $x = 0$
- The roots $x^2 - 2x + (1 - a)^2 = 0$ are
(a) $1 - a, -a - 1$ (b) $1 + a, 1 - a$ (c) $1, a$ (d) $1 - a, a$
- If a and b are the roots of equation $\frac{1}{x+1} + \frac{3}{5x+1} = \frac{5}{x+4}$, then $a + b =$
(a) $\frac{11}{17}$ (b) 1 (c) $-\frac{11}{17}$ (d) $\frac{6}{17}$
- Which of the following equations has two distinct real roots?
(a) $2x^2 - 3\sqrt{2}x + \frac{9}{4} = 0$ (b) $x^2 + x - 5 = 0$
(c) $x^2 + 3x + 2\sqrt{2} = 0$ (d) $5x^2 - 3x + 1 = 0$
- Find the value of p for which the quadratic equation $(p + 1)x^2 - 6(p + 1)x + 3(p + 9) = 0$ $p \neq -1$, has equal roots.
(a) -1 (b) 3 (c) -1 or 3 (d) -3
- What is / are the roots of $3x^2 = 6x$? [CBSE Practice Set - 1, 2003]
(a) Only 2 (b) Only 3 (c) 0 and 6 (d) 0 and 2
- Which of the following is a quadratic equation? [CBSE 2020]
(a) $x^2 + 2x + 1 = (4 - x)^2 + 3$ (b) $-2x^2 = (5 - x)\left(2x - \frac{2}{5}\right)$
(c) $(k + 1)x^2 + \frac{3}{2}x = 7$, where $k = -1$ (d) $x^3 - x^2 = (x - 1)^3$

10. The nature of roots of the quadratic equation $9x^2 - 6x - 2 = 0$ is : **[CBSE SQP, 2023]**
 (a) No real roots (b) 2 equal real roots (c) 2 distinct real roots (d) more than 2 real roots
11. Let p be a prime number. The quadratic equation having its roots as factors of p is :
 (a) $x^2 - px + p = 0$ (b) $x^2 - (p + 1)x + p = 0$ (c) $x^2 + (p + 1)x + p = 0$ (d) $x^2 - px + p + 1 = 0$
12. The roots of the equation $x^2 - 3x - m(m + 3) = 0$, where m is a constant, are :
 (a) $m, m + 3$ (b) $-m, m + 3$ (c) $m, -(m + 3)$ (d) $-m, -(m + 3)$
13. If the quadratic equation $mx^2 + 2x + m = 0$, has two equal roots, then the values of m are :
 (a) ± 1 (b) $0, 2$ (c) $0, 1$ (d) $-1, 0$
14. If $a - 3 = \frac{10}{a}$, then the values of a are :
 (a) $-2, 5$ (b) $5, 10$ (c) $2, 5$ (d) $2, -5$
15. The roots of $x + \frac{1}{x} = 2$ are :
 (a) $3, 2$ (b) $1, 1$ (c) $2, 2$ (d) $1, 2$

Part - 2 (Basic Subjective Type Questions)

- A shopkeeper buys a number of books for Rs 80. If he had bought 4 more books for the same amount, each book would have cost him Rs 1 less. How many books did he buy?
- A number consists of two digits whose product is 18. If 27 is added to the number, the number formed will have the digits in reverse order, when compared to the original number. Find the number.
- A thief runs with a uniform speed of 100 m/minute. After one minute, a policeman runs after the thief to catch him. He goes with a speed of 100m/minute in the first minute and increases his speed by 10m/minute every succeeding minute. After how many minutes the policeman will catch the thief?
- A train travels at a certain average speed for a distance of 63 km and then travels at a distance of 72 km at an average speed of 6 km/hr more than its original speed. If it takes 3 hours to complete journey, what is the original average speed?
- Find the value of k for which the equation $x^2 + k(2x + k - 1) + 2 = 0$ has real and equal roots.

[CBSE 2016]

- Find the value of m so that the quadratic equation $mx(5x - 6) + 9 = 0$ has two equal roots.

[CBSE Term - 2, SQP 2022]

- Find the roots of the following quadratic equations by the factorisation method :

(a) $2x^2 + \frac{5}{3}x - 2 = 0$

(b) $\frac{2}{5}x^2 - x - \frac{3}{5} = 0$

(c) $3\sqrt{2}x^2 - 5x - \sqrt{2} = 0$

[CBSE 2016]

(d) $3x^2 + 5\sqrt{5}x - 10 = 0$

[CBSE 2016]

(e) $21x^2 - 2x + \frac{1}{21} = 0$

[NCERT Exemplar]

8. Find the roots of the quadratic equations by using the quadratic formula in each of the following :

[NCERT Exemplar]

(a) $5x^2 + 13x + 8 = 0$

(b) $-3x^2 + 5x + 12 = 0$

(c) $-x^2 + 7x - 10 = 0$

(d) $x^2 + 2\sqrt{2}x - 6 = 0$

[CBSE 2015, 2016, 2017]

(e) $x^2 - 3\sqrt{5}x + 10 = 0$

[CBSE 2015, 2016, 2017]

(f) $\frac{1}{2}x^2 - \sqrt{11}x + 1 = 0$

[CBSE 2015, 2016, 2017]

9. If Ritu were younger by 5 years than what she really is, then the square of her age would have been 11 more than five times her present age. What is her present age? [CBSE Term-2, SQP 2022]

10. A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/h more than the first speed. If it takes 3 hours to complete the total journey, what is its first speed?

[Delhi Govt. QB 2022, CBSE 2015]



Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

- One of the two students, while solving a quadratic equation in x , copied the constant term incorrectly and got the roots 3 and 2. The other copied the constant term and coefficient of x^2 correctly as -6 and 1 respectively. The correct roots are
 - 3, -2
 - -3 , 2
 - -6 , -1
 - 6, -1
- If $\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = x + 2$, then $x =$
 - 3
 - 2
 - 4
 - None of these
- If $x = \sqrt{7 + 4\sqrt{3}}$, then $x + \frac{1}{x} =$
 - 4
 - 6
 - 3
 - 2
- If the roots of the equation $px^2 + 2qx + r = 0$ and $qx^2 - 2\sqrt{pr}x + q = 0$ be real, then
 - $p = q$
 - $q^2 = pr$
 - $p^2 = qr$
 - $r^2 = pq$
- The real roots of the equation $x^{2/3} + x^{1/3} - 2 = 0$ are
 - 1, 8
 - -1 , -8
 - -1 , 8
 - 1, -8
- Each root of $x^2 - bx + c = 0$ is decreased by 2. The resulting equation is $x^2 - 2x + 1 = 0$, then
 - $b = 6$, $c = 9$
 - $b = 3$, $c = 5$
 - $b = 2$, $c = -1$
 - $b = -4$, $c = 3$

7. Two distinct polynomials $f(x)$ and $g(x)$ are defined as follows : $f(x) = x^2 + ax + 2$; $g(x) = x^2 + 2x + a$. If the equations $f(x) = 0$ and $g(x) = 0$ have a common root, then the sum of the roots of the equation $f(x) + g(x) = 0$ is [KVPY]

(a) $-\frac{1}{2}$ (b) 0 (c) $\frac{1}{2}$ (d) 1.

8. If $x = \frac{3+\sqrt{5}}{2}$ and $y = x^3$, then y satisfies the quadratic equation [NTSE]

(a) $y^2 - 18y + 1 = 0$ (b) $y^2 + 18y + 1 = 0$
(c) $y^2 - 18y - 1 = 0$ (d) $y^2 + 18y - 1 = 0$

9. Read the following statement carefully and state 'T' for true and 'F' for false. [Olympiad]

(i) If $x = a$ is the solution of the equation $x^2 - (a + b)x + k = 0$, then the value of k is a/b .

(ii) If $a^2b^2x^2 + 2abx + 1 = 0$, then one of the solution of the equation is $x = -ab$.

(iii) The nature of roots of the equation is real and distinct.

	(i)	(ii)	(iii)
(a)	T	T	T
(b)	F	T	T
(c)	F	F	T
(d)	F	T	F

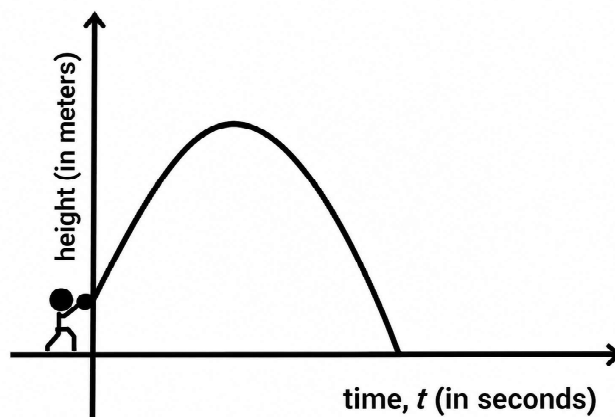
10. Find m .

$$3(2)^{2m} + 11(3)^m = 4$$

(a) -2 (b) 1 (c) -1 (d) 2

11. The graph below represents the path of a ball thrown by Ankush. The maximum height h , the ball reaches with respect to time t , is represented by the polynomial

$$h(t) = t^2 + \frac{19}{4}t + \frac{5}{4}$$



How long will it take for the ball to reach the ground?

(a) 5 seconds (b) 4 seconds (c) 10 seconds (d) 2 seconds

12. Three students were asked how they would verify their solution of a quadratic equation, $(x - 2)(x - 5) = 0$ Shown below are their responses. **[CBSE Competency Focused Practice Question 2022-23]**

Student 1 said, "In the first bracket, x must equal 2, and in the second bracket, x must equal 5. So, $(2 - 2)(5 - 5) = 0$."

Student 2 said, "In the first bracket, x must equal 2, but in the second bracket, x can have any real number value. For example, $(2 - 2)(3 - 5) = 0$ or $(2 - 2)(10 - 5) = 0$."

Student 3 said, "Both brackets should always have the same x value. So, x is either 2 or 5 in both brackets. For example, $(2 - 2)(2 - 5) = 0$ and $(5 - 2)(5 - 5) = 0$."

Whose response is correct?

- (a) only student 1 (b) only student 3
(c) only students 1 and 2 (d) all students 1, 2 and 3
13. A teacher asks three students to complete the following statement about the nature of the roots of a quadratic equation. **[CBSE Question Bank 2003]**

If $q^2 - 4pr > 0$, the roots of the quadratic equation $px^2 + qx + r = 0$ will be...

Zain answers, "always positive".

Vipul answers, "positive, if p, q and r are positive".

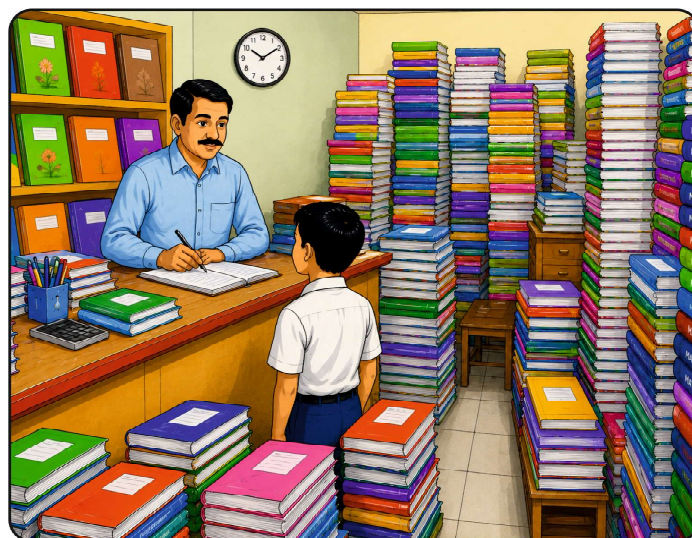
Suman answers, "negative, if p, q and r are positive".

Who answered correctly?

- (a) Zain (b) Vipul (c) Suman (d) none of them
14. For what values of k will quadratic equation $(2k + 1)x^2 + 2(k + 3)x + (k + 5) = 0$ have real and equal roots?

- (a) $\frac{-5 + \sqrt{41}}{2}$ (b) $\frac{-5 + \sqrt{41}}{2}$ or $\frac{-5 - \sqrt{41}}{2}$
(c) $\frac{-5 - \sqrt{41}}{2}$ (d) Cannot be determined

15. A shopkeeper buys a number of copies for a fixed amount. If he had bought few more copies for the same amount then each copy would have cost him less. To find out how many copies the shopkeeper had bought, Saksham formed a quadratic equation $(x^2 + 1)^2 - x^2 = 0$. **[NCERT Exemplar]**

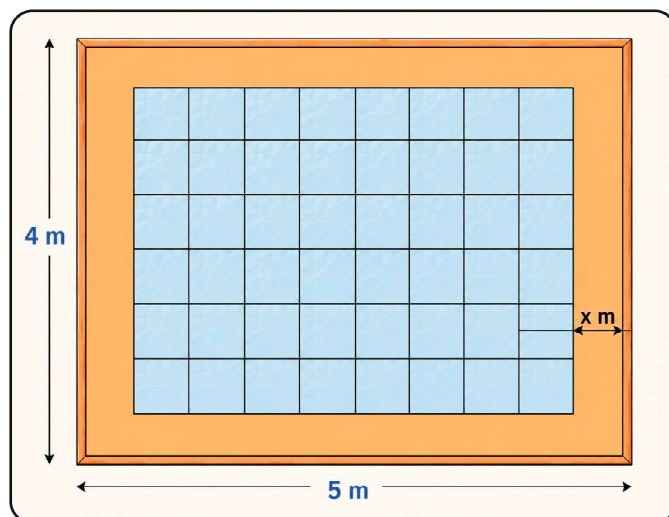


The quadratic equation has :

- (a) four real roots (b) two real roots (c) no real roots (d) one real root

Part - 2 (Advanced Subjective Type Questions)

- B takes 16 days less than A to do a piece of work. If both working together can do it in 15 days, in how many days will B alone complete the work?
- Two water taps together can fill a tank in $1\frac{7}{8}$ hours. The tap with longer diameter takes 2 hours less than the tap with smaller one to fill the tank separately. Find the time in which each tap can fill the tank separately.
- Two pipes together can fill a tank in $\frac{15}{8}$ hours. The pipe with larger diameter takes 2 hours less than the pipe with smaller diameter to fill the tank separately. Find the time in which each pipe can fill the tank separately.
- Out of a certain number of Saras birds one-fourth the number are moving about in lotus plants, $\frac{1}{9}$ th coupled with $\frac{1}{4}$ th as well as 7 times the square root of the number move on a hill, 56 birds remain in Vacula tree, what is the total number of birds?
- An equation has been given as $\frac{c}{x^2} + \frac{k}{x} = 1$.
Find the relation between c and k if
(i) x has real values (ii) x has no real values.
- Prove that the equation $x^2(a^2 + b^2) + 2x(ac + bd) + (c^2 + d^2) = 0$ has no real root, if $ad \neq bc$.
- Value of k for the quadratic equation $2x^2 + kx + 3 = 0$, so that it has two equal roots, is $\pm a\sqrt{6}$. The value of 'a' is
- Roots of the quadratic equation $36x^2 - 12ax + (a^2 - b^2) = 0$ are $\frac{a+b}{c}$ and $\frac{a-b}{c}$. The value of c is
- Some students planned to go for a picnic. The budget for food was Rs. 240. As four students failed to go, the cost of food for each student increased by Rs. 10. How many students had gone for the picnic?
- Ayush used the quadratic formula to solve a quadratic equation in y to get : $y = \frac{7 \pm \sqrt{169}}{10}$
Write a quadratic equation Ayush could have been solving. Show your steps.
- Find all the possible values of x for the following equation to be true : [CBSE Competency QB 2023]
$$\sqrt{(15 - 2x)} - x = 0$$
- If a and b are the roots of the equation $x^2 + ax - b = 0$, then find a and b.
- Solve the given quadratic equation for x : $9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0$. [CBSE 2016]
- Find two consecutive odd positive integers, sum of whose squares is 290. [CBSE SQP 2020]
- In flight of 2800 km, an aircraft was slowed down due to bad weather. Its average speed is reduced by 100 km/h and by doing so, the time of flight is increased by 30 minutes. Find the original duration of the flight. [CBSE 2024]
- A wall shown in figure measure 5 m in length and 4m in height. The outer portion of the wall of uniform width 'x' m will be painted and the central portion will be tiled. The total budget, including the tiles at Rs. 500 per m² and paint at Rs. 200 per m² is Rs. 5800. Find x such that the work is completed as per the budget. Show your work. [CBSE Question Bank 2023]



17. To fill a swimming pool two pipes are used. If the pipe of larger diameter used for 4 hours and the pipe of smaller diameter for 9 hours, only half of the pool can be filled. Find, how long it would take for each pipe to fill the pool separately, if the pipe of smaller diameter takes 10 hours more than the pipe of larger diameter to fill the pool?
[CBSE SQP Std. 2022]
18. Amy (A) and Benny (B), working together, can plant a magical rainbow flower garden in 6 days. Amy (A) with her fast planting skills, takes 5 days less than Benny (B) to finish planting the work. In how many days can Benny (B) do the same work alone?
19. Rs. 9,000 were divided equally among a certain number of persons. Had there been 20 more persons, each would have got Rs. 160 less. Find the original number of persons.
[CBSE 2020]

Part - 3 (Advanced Miscellaneous Questions)

1. Match the equations in column I with nature of roots in column II and choose the correct option.

Column-I	Column-II
(A) $x^2 - 3x + 11 = 0$	(i) Two equal roots
(B) $x^2 - 5x + 4 = 0$	(ii) No real roots
(C) $x^2 - 8x + 16 = 0$	(iii) Two distinct real roots
(D) $x^2 - 6x + 9 = 0$	(iv) Two equal roots

- (a) A-ii, B-iv, C-i, D-iii
(c) A-iv, B-i, C-ii, D-iii

- (b) A-ii, B-iii, C-iv, D-i
(d) A-iii, B-iv, C-i, D-ii

2. Match the equations given in Column - I with their roots in Column - II and choose the correct option.

Column-I	Column-II
(A) $4x^2 + 4\sqrt{3}x + 3 = 0$	(i) $x = 6, 2$
(B) $3x^2 + 5x - 7 = 0$	(ii) $x = \frac{-\sqrt{3}}{2}$
(C) $x^2 + 8x + 12 = 0$	(iii) $x = \frac{1}{2}, 3$
(D) $2x^2 + 5x - 3 = 0$	(iv) $x = \frac{-5 \pm \sqrt{109}}{6}$

- (a) A-iii, B-iv, C-i, D-ii
(c) A-ii, B-iv, C-iii, D-i

- (b) A-iv, B-iii, C-ii, D-i
(d) A-ii, B-iv, C-i, D-iii

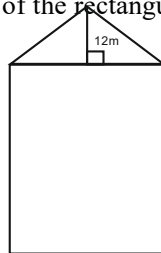
- **Directions :** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:
 (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
 (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
 (c) Assertion (A) is true but reason (R) is false.
 (d) Assertion (A) is false but reason (R) is true.
- 3. **Assertion (A) :** When a marble is dropped from an initial height, d metres, with an initial speed, v m/s, the height of the marble at time t is represented by $h(t) = vt - 2t^2 + d$. Now, when a marble is dropped from a height of 48 m with an initial speed of 10 m/s, the time taken by it to hit the ground is 8 seconds.
Reason (R) : In a quadratic equation, the highest degree of the variable is 2.
- 4. **Assertion (A) :** $(x + 2)^3 = 2x(x^2 - 1)$ is not a quadratic equation.
Reason (R) : A quadratic equation in the variable x is an equation of the form $ax^2 + bx + c = 0$ where a, b, c are any real numbers.
- 5. **Assertion (A) :** $\sqrt{11} + 3, \sqrt{11} - 3$ and 0 are the roots of the quadratic equation $\frac{1}{2}x^2 - \sqrt{11}x + 1 = 0$
Reason (R) : A quadratic equation can have atmost two real roots.
- 6. **Assertion (A) :** It is possible to design a rectangular mango grove whose length is twice its breadth, and the area is 800m^2 .
Reason (R) : In quadratic equation $ax^2 + bx + c = 0$, if $b^2 - 4ac = 0$ then the roots of the equation are real and distinct.
- 7. **Assertion (A) :** It is possible that the sum of the ages of two friends is 20 years and four years ago, the product of their ages in years was 48.
Reason (R) : In quadratic equation $ax^2 + bx + c = 0$, if $b^2 - 4ac > 0$, then the roots of the equation are real and distinct.



Exercise-3

(Case Based Questions)

1. A rectangular park is to be designed whose breadth is 4 m less than its length. Its area is to be 80 square metres more than the area of a park that has already been made in the shape of an isosceles triangle with its base as the breadth of the rectangular park and of altitude 12 m shown in figure.



Answer the following questions.

- | | | | | |
|--|--------------------|---------------------|---------------------|--------------------|
| (i) Find the breadth of the park. | (a) 8 m | (b) 10 m | (c) 80 m | (d) 2 m |
| (ii) Find the length of the park. | (a) 8 m | (b) 10 m | (c) 14 m | (d) 6 m |
| (iii) Find the area of rectangular part of park. | (a) 80m^2 | (b) 140m^2 | (c) 200m^2 | (d) 60m^2 |
| (iv) Find the area of triangular part of park. | (a) 60m^2 | (b) 140m^2 | (c) 50m^2 | (d) 80m^2 |

- (v) Find the area of park.
 (a) 200 m^2 (b) 140 m^2 (c) 60 m^2 (d) 160 m^2
2. Let us consider a quadratic equation $x^2 + \lambda x + \lambda + 1.25 = 0$, where λ is a constant. The value of λ such that the above quadratic equation has
- (i) Two distinct roots
 (a) $\lambda < 5$ (b) $\lambda > -1$ (c) $\lambda > 5$ or $\lambda < -1$ (d) None of these
- (ii) Two coincident roots
 (a) $\lambda < 5$ or $\lambda = -1$ (b) $\lambda = 1$ or $\lambda = 5$ (c) $\lambda = -5$ or $\lambda = 1$ (d) None of these
3. The tradition of pottery making in India is very old. In fact, it is older than Indus Valley Civilization. The shaping and baking of clay articles has continued through the ages. The picture of a potter is shown below :



A potter makes a certain number of pottery articles in a day. It was observed on a particular day the cost of production of each article (in Rs.) was one more than twice the number of articles produced on that day. The total cost of production on that day was Rs. 210.

The number of articles produced on that day is x .

- (i) Find the cost of each article in terms of x .
 (ii) Form a quadratic equation representing the given situation in terms of x .
 (iii) Find the number of articles produced and the cost of each article.

OR

- (iii) If the total cost of production on that day was Rs. 300, find the number of articles produced.
4. The two gears shown in the figure resemble two circles which touch each other externally. The sum of their areas is $130\pi \text{ sq. cm}$ and distance between their centres is 14 cm.



- (i) If the radius of one circle to be $x \text{ cm}$, then find the quadratic equation for the situation described above.
 (ii) Determine the nature of roots of the quadratic equation of given situation.
 (iii) Find the radii of the two circles.

Answer Key



Exercise-1

Part - 1 (Basic Multiple Choice Questions)

- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| (1) B | (2) A | (3) B | (4) B | (5) D | (6) B |
| (7) B | (8) D | (9) D | (10) B | (11) B | (12) B |
| (13) A | (14) A | (15) B | | | |

Part - 2 (Basic Subjective Type Questions)

- (1) Number of books bought = 16
- (2) The required numbers are -63 and 36
- (3) The policeman catches the thief after 5 minute
- (4) Original average speed = 42km/hr.
- (5) $k = 2$
- (6) The value of m is 5.
- (7) (a) Roots : $x = -\frac{3}{2}, \frac{2}{3}$ (b) Roots : $x = -\frac{1}{2}, 3$
- (c) Roots : $x = \sqrt{2}, -\frac{\sqrt{2}}{6}$ (d) Roots : $x = \frac{\sqrt{5}}{3}, -2\sqrt{5}$
- (e) Roots : $x = \frac{1}{21}, \frac{1}{21}$
- (8) (a) Roots : $x = -1, -1.6$ (b) Roots : $x = -\frac{4}{3}, 3$
- (c) Roots : $x = 2, 5$ (d) Roots : $x = \sqrt{2}, -3\sqrt{2}$
- (e) Roots : $x = 2\sqrt{5}, \sqrt{5}$ (f) Roots : $x = \sqrt{11} + 3, \sqrt{11} - 3$
- (9) Ritu's present age is 14 years. (10) The original average speed of train is 36km/h.



Exercise-2

Part - 1 (Advanced Basic Multiple Choice Questions)

- (1) D (2) C (3) A (4) B (5) D (6) A
 (7) C (8) A (9) C (10) C (11) A (12) B
 (13) C (14) B (15) C

Part - 2 (Advanced Subjective Type Questions)

- (1) Number of days that are taken by B to finish the work alone = $x - 16 = 40 - 16 = 24$
 (2) Time taken by tap with smaller diameter = 5 hours and time taken by tap with longer diameter = 3 hours.
 (3) Two pipe take 5 hours and 3 hours respectively.
 (4) He has Rs 54 and there were 5 persons The total number of birds are 576.
 (5) $c \geq \frac{-k^2}{4}$
 (6) The given equation has no real roots.
 (7) $a = 2$
 (8) $c = 6$
 (9) The number of students who went for the picnic = $12 - 4 = 8$.
 (10) Ayush used the quadratic formula to solve a quadratic equation in y to get $y = \frac{7 \pm \sqrt{169}}{10}$
 (11) The value of x are 3 and -5
 (12) $a = -1$ and $b = 2$.
 (13) $x = \frac{2a + b}{3}$ or $x = \frac{a + 2b}{3}$.
 (14) The odd positive integers are 11 and 13.
 (15) Time = 3 hr 30 min.
 (16) x is 1m.
 (17) Larger pipe would alone fill the tank in 20 hours and smaller pipe would fill the tank alone in 30 hours
 (18) Benny (B) takes 15 days to complete the work alone.
 (19) The original number of persons is 25.

Part - 3 (Advanced Miscellaneous Questions)

- (1) A (2) D (3) B (4) C (5) D (6) D
(7) D



Exercise-3

(Case Based Questions)

- (1) (i) 10m (ii) 14m (iii) 140 m² (iv) 60 m² (v) 200m²

- (2) (i) $\lambda > 5$ or $\lambda < -1$ (ii) $\lambda < 5$ or $\lambda = -1$

- (3) (i) The cost of each article = $1 + 2x$

(ii) $2x^2 + x - 210 = 0$

- (iii) The number of articles cannot be negative $\Rightarrow x = 10$

Cost of each article = $1 + 2(10) = 21$

OR

- (iii) The number of articles cannot be negative $\Rightarrow x = 12$

- (4) (i) $x^2 - 14x + 33 = 0$

- (ii) Since, $D > 0$

$\therefore$ The roots of given quadratic equation are real and distinct.

- (iii) The radii of the two circles are 3cm and 11cm.